Biologic Adherence: COVID-19 Vaccine Timing Dilemma

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BACKGROUND

- A lack of evidence-based recommendations regarding the timing of administration of immunomodulary medications with respect to COVID-19 vaccination led to inconsistent provider recommendations and patient confusion.
- In rheumatology, the American College of Rheumatology (ACR) published guidelines[1] to standardize recommendations and to ensure medication adherence and optimal immune response to the COVID-19 vaccine. Guidance varied based on drug class, perpetuating confusion on whether to administer or not administer medication.
- Health system specialty pharmacists are in a unique position to provide appropriate education and manage medication adherence in this patient population.

OBJECTIVES

- The primary objective is to determine the impact of pharmacist education on patient alignment with ACR guidelines regarding appropriateness of biologic DMARD dosing with respect to COVID-19 vaccination
- The secondary objective is to measure pharmacist impact on COVID-19 vaccination rates

METHODS

Study Design

This was a single-center, observational study. This was conducted as a chart review of autoimmune patients established on a biologic DMARD and eligible for COVID-19 vaccination. As part of the clinical protocol, pharmacists educate patients on the importance of vaccinations. Pharmacists are expected to provide education to combat misinformation, ensure alignment with published guidelines and assist patients with scheduling of vaccinations.

INCLUSION CRITERIA	EXCLUSION CRITERIA
• Patients enrolled in clinical management at the Parkview Specialty Pharmacy Rheumatology clinic who had not yet been vaccinated against COVID-19 • Patients established on a biologic DMARD • Patients with a routine assessment scheduled to be completed within the 6 months observed	• Patients within Rheumatology clinic on a biologic DMARD who are not clinically enrolled in Parkview Specialty Pharmacy clinical services • Patients that did not wish to discuss COVID-19 with the pharmacist • Patients who had already been vaccinated against COVID-19

DATA COLLECTION AND ENDPOINTS

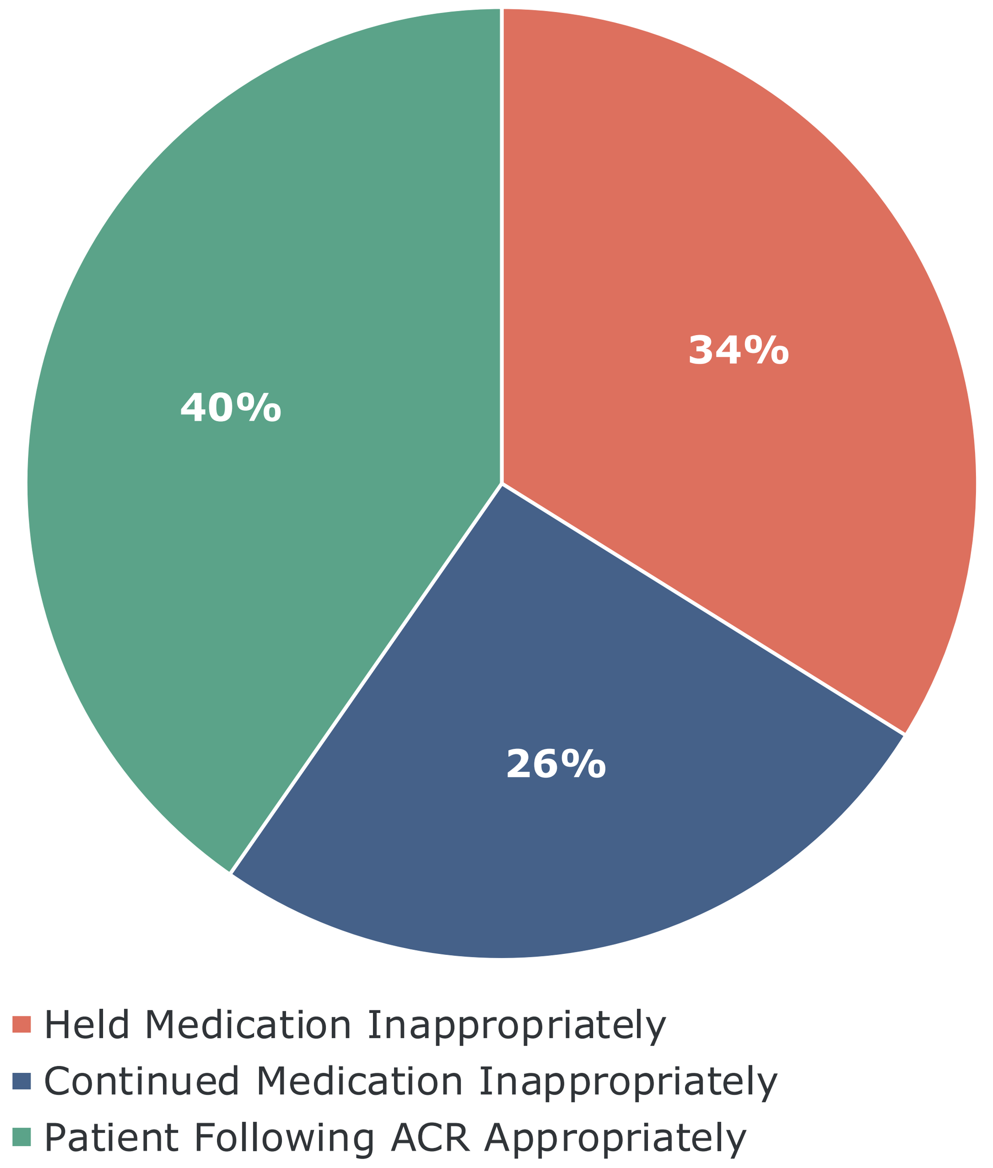
During counseling sessions, patients were asked "Have you given any consideration on how to take your biologic medication with regard to COVID-19 vaccination?" Pharmacists documented the variances from published ACR recommendations, and counseled patients on the proper timing of doses as appropriate.

PRIMARY ENDPOINTS	SECONDARY ENDPOINT
• Prevented unnecessary hold: Patients intended to hold medication inappropriately. Pharmacists counseled on appropriate continuation of medication to prevent lost days of therapy. • Recommended therapy hold: Patient intended to continue medication throughout COVID-19 vaccination series. Pharmacist counseled on appropriate holding of medication based on ACR guidelines to ensure optimal vaccination response and protection against COVID-19.	• Pharmacist impact on vaccination rate: Total number of patients that accepted pharmacist recommendation who previously declined COVID-19 vaccination.

RESULTS

Primary Endpoint

Pharmacist Intervention Analysis (n=62)



21 (34%) would have held medication inappropriately

16 (26%) would have continued medication inappropriately

Pharmacist interventions prevented 60% of patients who received the vaccine from incorrectly holding or continuing medication

12 patients previously refused vaccination but will be vaccinated after pharmacist intervention

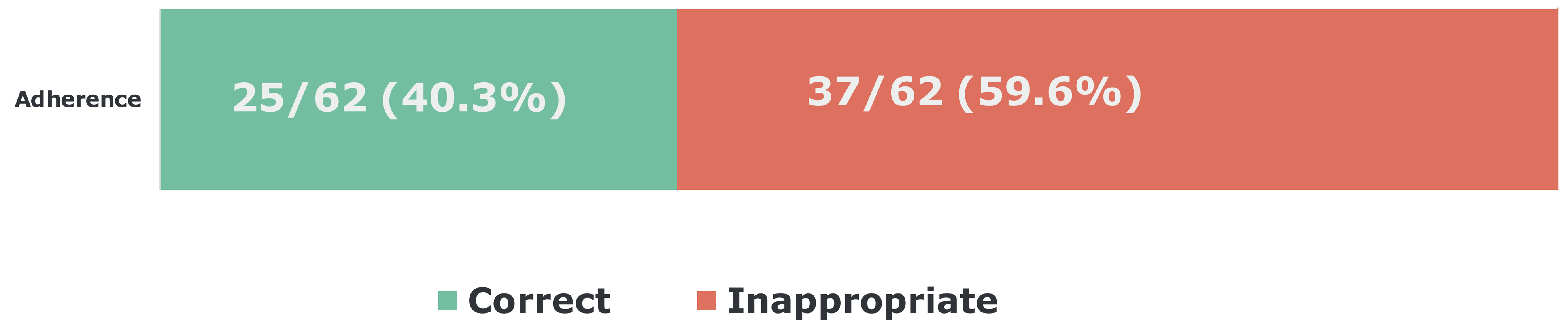
Secondary Endpoint: Increasing Vaccination Rate

Total Patients	Assessments completed	82
	Declined vaccination	20
Vaccination Recommendation Accepted	Planning on getting vaccinated	62
	New commit to vaccine	12
	Pharmacist assisted scheduling	4

DISCUSSION AND CONCLUSIONS

Alignment with ACR Guidelines

Three out of five patients would have incorrectly administered medication.



Discussion and Impact

- Pharmacist intervention resulted in a positive impact on patient alignment with ACR dosing guidelines.
- Analysis found that 60% of patients would have inappropriately dosed biologic DMARD without pharmacist intervention.
- Assuming a minimum hold of 14 days of therapy missed, on average, with each vaccination, pharmacist education prevented approximately 588 days of missed therapy in aggregate.
- Twenty-eight days missed therapy results in approximate loss of nearly 8% total annualized PDC per patient.
- Due to pharmacist intervention, 16 patients improved vaccine efficacy through holding DMARD correctly. Median cost of a COVID-19 infection per patient is estimated to be $34,662 per infection, resulting in a potential savings of up to $550,000 in overall healthcare spend.[2]
- As of August 31, 2021, only half of the residents of Indiana are fully vaccinated and cases are on the rise with nearly 2,000 new cases reported daily.[3]

REFERENCES

1. Curtis, Jeffery et al. American College of Rheumatology Guidance for COVID-19 Vaccination in Patients With Rheumatic and Musculoskeletal Diseases: Version 1. Arthritis and Rheumatology March 2021, 73 (7) 1093-1107
2. Hackett, Mallory Hospitalized care for COVID-19 averages $34,662 to $45,683, varying by age. Healthcare Finance. July 2020
3. Indiana State Department of Health Immunization Dashboard reporting as of 8/13/2021